CENTRAL BOARD OF SECONDARY EDUCATION

JAWAHAR NAVODAYA VIDYALAYA

TARIKHET ALMORA UTTARAKHAND 263663

COMPUTER SCIENCE (083)

PROJECT FILE

On

SOCIAL MEDIA SURVEY SYSTEM

USING PYTHON'S TKINTER LIBRARY AND MySQL

SUBMITTED BY :

S. No	Name	CBSE Roll
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4.		

CLASS : XII
SECTION : SCIENCE

INTERNAL EXAMINER (PANKAJ BHILINGWAL)	EXTERNAL EXAMINER	PRINCIPAL (PANKAJ KUMAR SRIVASTAVA)

ACKNOWLEDGEMENT

*I wish to express my deep sense of gratitude and indebtedness to our learned teacher **Pankaj Bhilingwal**, PGT Computer Science, J.N.V Tarikhet Almora for his invaluable help, advice and guidance in the preparation of this project file.*

*I am also greatly indebted to our principal **Pankaj Kumar Srivastava** and school authorities for providing me with the facilities and requisite laboratory conditions for making this project file.*

I also extend my thanks to a number of teachers, my classmates and friends who helped me to complete this project file successfully.

NAME :

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CLASS : XII
SECTION : SCIENCE

CERTIFICATE

This is to certify that,

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Student of Class XII, J.N.V Tarikhet Almora has completed the PROJECT FILE during the academic year 2026-27 towards partial fulfilment of credit for the Computer Science Project evaluation of CBSE and submitted satisfactory report, as compiled in the following pages, under my supervision.

Internal Examiner
Signature

External Examiner
Signature

PREFACE

In the field of Computer Science, the term PROJECT refers to a Computerization Work that is completely performed by using a computer. Today when every field is going to be computerized means that all task of any field now a days done by using computer and their appropriate computer programs, then to do every field's different-different works, today different programs or software's are available in the form of Project.

Learning Python & MySQL in schools is a part of our syllabus, and according to the syllabus worked out by the CBSE, to which my school is affiliated, I enjoyed doing project work for the session 2026-27. I learned a lot, especially learning by doing was a great experience. The report is a summary of what I observed and learnt in school. My humble thanks are going to all the concerned related to my project.

I also feel that it is the right time to thank my teacher, who guided me during my project. I also feel that it is the right time to thank my parents, friends and my project partners for their constant assistance, cooperation extended by them time to time.

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About Project

Social Media Survey System

The Social Media Survey System is a user-friendly application developed in Python, leveraging the Tkinter library for the graphical user interface (GUI) and MySQL as the database backend. This project serves as a survey tool to collect and manage information about users' social media preferences.

Project Overview

Objective:

The main objective of the Social Media Survey System is to gather insights into users' social media habits. By utilizing a Tkinter-based GUI, users can easily input their name, select the social media platforms they use, specify the platform they spend the most and least time on, and provide their date of birth. The collected data is then stored in a MySQL database for efficient retrieval and analysis.

Key Features:

1. **Data Collection:** Users can input their name, select social media platforms used, and specify the most and least used platforms.
2. **Data Storage:** Utilizes MySQL to store survey responses, ensuring organized and accessible data management.
3. **Data Retrieval:** Offers functionalities to view survey information and user details, aiding in data analysis.
4. **Data Deletion:** Provides a secure and user-friendly way to delete specific survey entries based on user ID.

Technology Stack

- ✓ **Programming Language:** Python
- ✓ **GUI Library:** Tkinter
- ✓ **Database:** MySQL
- ✓ **Modules Used:** tkinter**,** mysql.connector

Description:

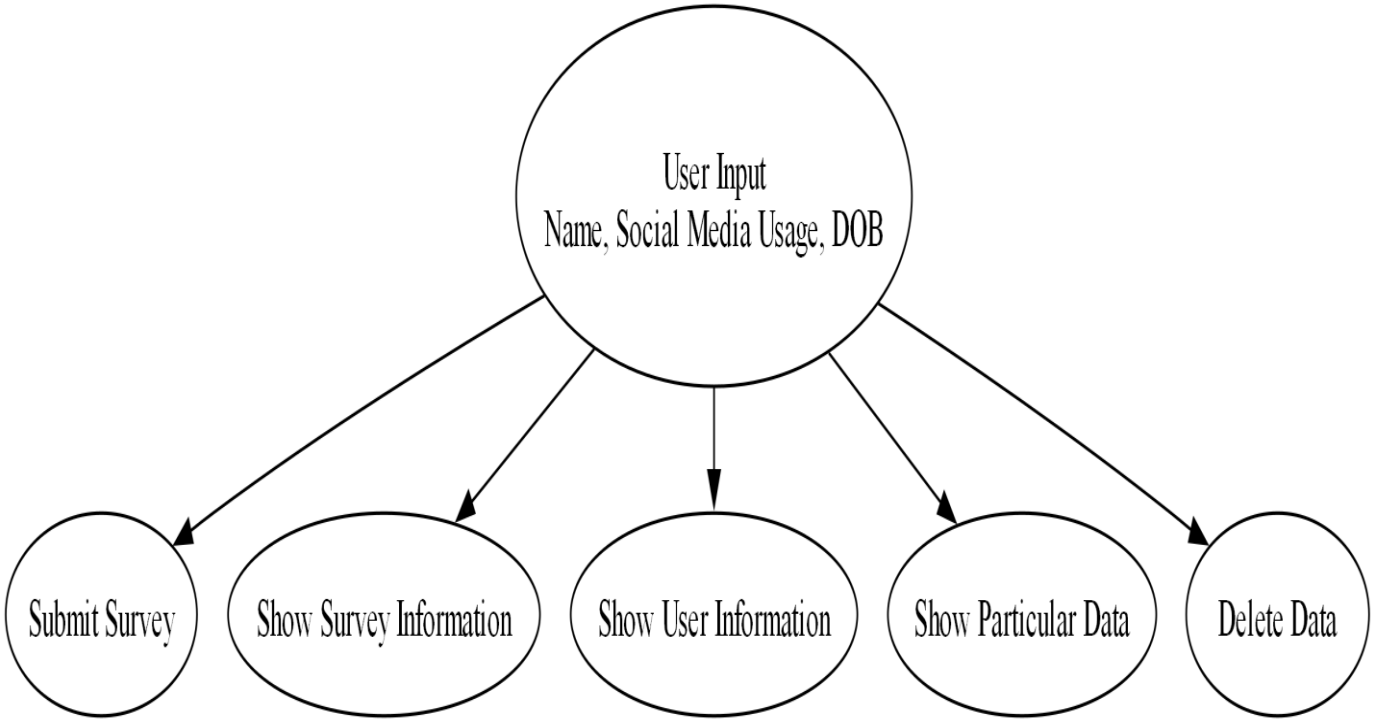
The project relies on Tkinter for creating an interactive GUI, allowing users to input survey data seamlessly. The mysql.connector module facilitates the connection to a MySQL database, where user survey responses and personal details are stored in separate tables.

How to Run the Program

1. Ensure Python is installed on your system.
2. Install the mysql-connector module using pip install mysql-connector.
3. Set up a MySQL server locally or remotely and update the connection details in the Python script.
4. Execute the main program file to launch the GUI.
5. Input survey data and explore functionalities such as viewing, deleting, and analysing collected information.

This Social Media Survey System streamlines the process of data collection and analysis, providing valuable insights into users' social media preferences.

Flow Chart


User Input
Name, Social Media Usage, DOB
Submit Survey
Show Survey Information
Show User Information
Show Particular Data
Delete Data

Project Code

```
from tkinter import *
from tkinter import ttk
import mysql.connector


root = Tk()
root.configure(bg='#ffeae0')


# Connect to MySQL database
my_db = mysql.connector.connect(host='localhost', user='root', password='admin')
my_db_cursor = my_db.cursor()
my_db_cursor.execute('CREATE DATABASE IF NOT EXISTS survey')
my_db.commit()


my_db1 = mysql.connector.connect(host='localhost', user='root', password='admin',
database='survey')
cur = my_db1.cursor()
cur.execute('''CREATE TABLE IF NOT EXISTS survey_info
        (
        id INT PRIMARY KEY,
        facebook VARCHAR(20),
        instagram VARCHAR(20),
        telegram VARCHAR(20),
        twitter VARCHAR(20),
        youtube VARCHAR(20),
        reddit VARCHAR(20),
        snapchat VARCHAR(20),
        whatsapp VARCHAR(20),
        quora VARCHAR(20),
        tiktok VARCHAR(20),
        most_time_spent VARCHAR(20),
```

```
        least_time_used VARCHAR(20)
        );'")

cur.execute('''CREATE TABLE IF NOT EXISTS user_info
        (
        roll INT PRIMARY KEY,
        name VARCHAR(255),
        date_of_birth DATE
        );''')



def submit():
    my_db1 = mysql.connector.connect(host='localhost', user='root', password='admin',
database='survey')
    cur = my_db1.cursor()

    # Get the values of the checkboxes
    checkbox_values = ['Yes' if var.get() else 'No' for var in checkboxes]

    # Create a tuple with checkbox values and other data
    survey_values = (
        roll_entry.get(),checkbox_values[0], checkbox_values[1], checkbox_values[2],
checkbox_values[3],
        checkbox_values[4], checkbox_values[5], checkbox_values[6], checkbox_values[7],
        checkbox_values[8], checkbox_values[9], clicked1.get(), clicked2.get()
    )

    b1 = 'INSERT INTO
survey_info(id,facebook,instagram,telegram,twitter,youtube,reddit,snapchat,whatsapp,quora,tiktok
,most_time_spent,least_time_used) VALUES(%s,%s,%s,%s,%s,%s,%s,%s,%s,%s,%s,%s,%s)'
    cur.execute(b1, survey_values)
```

```
    b2 = 'INSERT INTO user_info(roll, name, date_of_birth) VALUES(%s, %s, %s)'
    user_values = (roll_entry.get(), name_entry.get(), dob_entry.get())
    cur.execute(b2, user_values)

    name_entry.delete(0, END)
    roll_entry.delete(0, END)
    for var in checkboxes:
        var.set('No')  # Reset checkboxes to 'No'
    clicked1.set('select')
    clicked2.set('select')
    dob_entry.delete(0, END)
    dob_entry.insert(0, 'YYYY-MM-DD')
    my_db1.commit()
    Label(root, text='Data Recorded', bg='#ffeae0').grid(columnspan=2)


def show_survey():
    A = Tk()
    A.configure(bg='#ffeae0')
    A.title("Survey Information")


    tree = ttk.Treeview(A)
    tree["columns"] = ("Id", "Facebook", "Instagram", "Telegram", "Twitter", "YouTube", "Reddit",
"Snapchat",
                       "WhatsApp", "Quora", "TikTok", "Most Time Spent", "Least Time Used")


    # Define column headings
    tree.heading("Id", text="Id")
    tree.heading("Facebook", text="Facebook")
    tree.heading("Instagram", text="Instagram")
    tree.heading("Telegram", text="Telegram")
```

```
tree.heading("Twitter", text="Twitter")
tree.heading("YouTube", text="YouTube")
tree.heading("Reddit", text="Reddit")
tree.heading("Snapchat", text="Snapchat")
tree.heading("WhatsApp", text="WhatsApp")
tree.heading("Quora", text="Quora")
tree.heading("TikTok", text="TikTok")
tree.heading("Most Time Spent", text="Most Time Spent")
tree.heading("Least Time Used", text="Least Time Used")

# Set column widths statically
tree.column("Id", width=50)
tree.column("Facebook", width=60)
tree.column("Instagram", width=70)
tree.column("Telegram", width=60)
tree.column("Twitter", width=60)
tree.column("YouTube", width=60)
tree.column("Reddit", width=60)
tree.column("Snapchat", width=70)
tree.column("WhatsApp", width=80)
tree.column("Quora", width=60)
tree.column("TikTok", width=60)
tree.column("Most Time Spent", width=120)
tree.column("Least Time Used", width=120)
tree["show"] = "headings"  # Hide the default first column
# Populate treeview with data
cur.execute('SELECT * FROM survey_info')
for row in cur:
    tree.insert("", "end", values=row)
```

```
    # Add treeview to the window
    tree.grid(row=1, column=0, padx=1, pady=1, columnspan=2, sticky="nsew")

    # Add a vertical scrollbar
    scrollbar = ttk.Scrollbar(A, orient="vertical", command=tree.yview)
    scrollbar.grid(row=1, column=2, sticky="ns")
    tree.configure(yscrollcommand=scrollbar.set)

    A.mainloop()



def sort_treeview(tree, col, reverse):
    items = [(tree.set(k, col), k) for k in tree.get_children('')]
    items.sort(reverse=reverse)
    # rearrange items in sorted positions
    for index, (val, k) in enumerate(items):
        tree.move(k, '', index)
    tree.heading(col, command=lambda: sort_treeview(tree, col, not reverse))



def show_user_info():
    B = Tk()
    B.configure(bg='#ffeae0')
    B.title("User Information")

    tree = ttk.Treeview(B)
    tree["columns"] = ("Id","Name", "Date of Birth")

    # Define column headings with centered text
    for col in tree["columns"]:
```

```
        tree.heading(col, text=col, anchor="center")

    # Set column widths
    for col in tree["columns"]:
        tree.column(col, width=100, anchor="center")  # Adjust the width as needed
        tree["show"] = "headings"  # Hide the default first column

    # Populate treeview with data
    cur.execute('SELECT * FROM user_info')
    for row in cur:
        tree.insert("", "end", values=row)

    # Add treeview to the window
    tree.grid(row=1, column=0, padx=10, pady=10, columnspan=2, sticky="nsew")

    # Add a vertical scrollbar
    scrollbar = ttk.Scrollbar(B, orient="vertical", command=tree.yview)
    scrollbar.grid(row=1, column=2, sticky="ns")
    tree.configure(yscrollcommand=scrollbar.set)

    B.mainloop()

def delete_data():
    if vall.get() == '':
        Label(C, text='Enter id', bg='#ffeae0').grid(columnspan=2)
    else:
        s = (int(vall.get()),)
        b2 = 'SELECT * FROM  user_info where roll=%s'
        cur.execute(b2, s)
        ch = ''
```

```
        for a in cur:
            ch = ch + str(a[0])
        if ch == '':
            Label(C, text=f'No data exists where id = {vall.get()}', bg='#ffeae0').grid(columnspan=2)
            vall.delete(0, END)
        elif ch != '':
            s = (int(vall.get()),)
            b = 'DELETE FROM survey_info where id=%s'
            cur.execute(b, s)
            b1 = 'DELETE FROM user_info where roll=%s'
            cur.execute(b1, s)
            Label(C, text='Data deleted', bg='#ffeae0').grid(columnspan=2)
            vall.delete(0, END)
            my_db1.commit()


def delete_data_window():
    global vall
    global C
    C = Tk()
    C.configure(bg='#ffeae0')
    Label(C, text='DELETE WINDOW', bg='#ffeae0').grid(row=0, column=0, columnspan=2)
    Label(C, text='Enter Id of data to be deleted', bg='#ffeae0').grid(row=1, column=0)
    vall = Entry(C)
    vall.grid(row=1, column=1)
    Button(C, text='DELETE DATA', command=delete_data).grid(row=2, column=0,
columnspan=2, pady=10, padx=10, ipadx=100)
    C.mainloop()



def show_data():
```

```
    if valll.get() == '':
        Label(D, text='Enter id', bg='#ffeae0').grid(columnspan=2)
    else:
        s = (int(valll.get()),)
        b2 = 'SELECT * FROM user_info where roll=%s'
        cur.execute(b2, s)
        ch = ''
        for a in cur:
            ch = ch + str(a[0])
        if ch == '':
            Label(D, text=f'No data exists where id = {valll.get()}', bg='#ffeae0').grid(columnspan=2)
            valll.delete(0, END)
        elif ch != '':
            s = (int(valll.get()),)
            b = 'SELECT * FROM survey_info where id=%s'
            cur.execute(b, s)

            # Fetch all rows from the result set
            rows = cur.fetchall()

            E = Tk()
            E.configure(bg='#ffeae0')
            E.title("Data Information")

            tree = ttk.Treeview(E)
            tree["columns"] = ("Survey ID", "Facebook", "Instagram", "Telegram", "Twitter",
"YouTube", "Reddit",
                               "Snapchat", "WhatsApp", "Quora", "TikTok", "Most Time Spent", "Least Time
Used")

            # Configure column headings
```

```
        for col in tree["columns"]:
            tree.heading(col, text=col, anchor='center')
            tree.column(col, anchor='center', width=60)  # Adjust width as needed
            tree["show"] = "headings"  # Hide the default first column
        # Populate treeview with data using the fetched rows
        for row in rows:
            tree.insert("", "end", values=row)

        # Add a scrollbar to the treeview
        scrollbar = ttk.Scrollbar(E, orient="vertical", command=tree.yview)
        scrollbar.pack(side="right", fill="y")
        tree.configure(yscrollcommand=scrollbar.set)

        tree.pack(expand=YES, fill=BOTH)
        valll.delete(0, END)
        my_db1.commit()
        E.mainloop()


def show_particular_data_window():
    global D
    global valll
    D = Tk()
    D.configure(bg='#ffeae0')
    Label(D, text='VIEW WINDOW', bg='#ffeae0').grid(row=0, column=0, columnspan=2)
    Label(D, text='Enter id of data to view', bg='#ffeae0').grid(row=1, column=0)
    valll = Entry(D)
    valll.grid(row=1, column=1)
    Button(D, text='VIEW', command=show_data).grid(row=2, columnspan=2)
    D.mainloop()
```

```
Label(root, text='Name', bg='#ffeae0').grid(row=0, column=0)
Label(root, text='Roll', bg='#ffeae0').grid(row=0, column=2)
Label(root, text='Select the social media sites you use:', bg='#ffeae0').grid(row=1, column=0)

checkboxes = []
options = [
    'facebook', 'instagram', 'telegram', 'twitter', 'youtube', 'reddit', 'snapchat', 'whatsapp', 'quora', 'tiktok'
]

# Create Checkbuttons in a grid layout
row_num = 2
col_num = 0
for option in options:
    var = BooleanVar()
    checkbox = Checkbutton(root, text=option, variable=var, onvalue=True, offvalue=False, justify='left', bg='#ffeae0')
    checkbox.deselect()
    checkbox.grid(row=row_num, column=col_num, sticky=W, padx=5, pady=5)
    checkboxes.append(var)

    col_num += 1
    if col_num == 5:  # Adjust the number of columns as needed
        col_num = 0
        row_num += 1


name_entry = Entry(root)
name_entry.grid(row=0, column=1)
```

```
name_entry.insert(0, '')

roll_entry = Entry(root)
roll_entry.grid(row=0, column=3)
roll_entry.insert(0, '')



Label(root, text='which social media site you use most', bg='#ffeae0').grid(row=12, column=0)
Label(root, text='which social media site you use least', bg='#ffeae0').grid(row=13, column=0)

clicked1 = StringVar()
clicked1.set('select')
drop1 = OptionMenu(root, clicked1, *options)
drop1.grid(row=12, column=1)

clicked2 = StringVar()
clicked2.set('select')
drop2 = OptionMenu(root, clicked2, *options)
drop2.grid(row=13, column=1)

Label(root, text='Date of birth', bg='#ffeae0').grid(row=14, column=0)

dob_entry = Entry(root)
dob_entry.insert(0, 'YYYY-MM-DD')
dob_entry.grid(row=14, column=1)

Button(root, text='SUBMIT', command=submit).grid(row=15, column=0, columnspan=2,
pady=10, padx=10, ipadx=100)
my_db1.commit()
```

```
Button(root, text='SHOW SURVEY INFO', command=show_survey).grid(row=16, column=0, columnspan=2, pady=10, padx=10, ipadx=100)

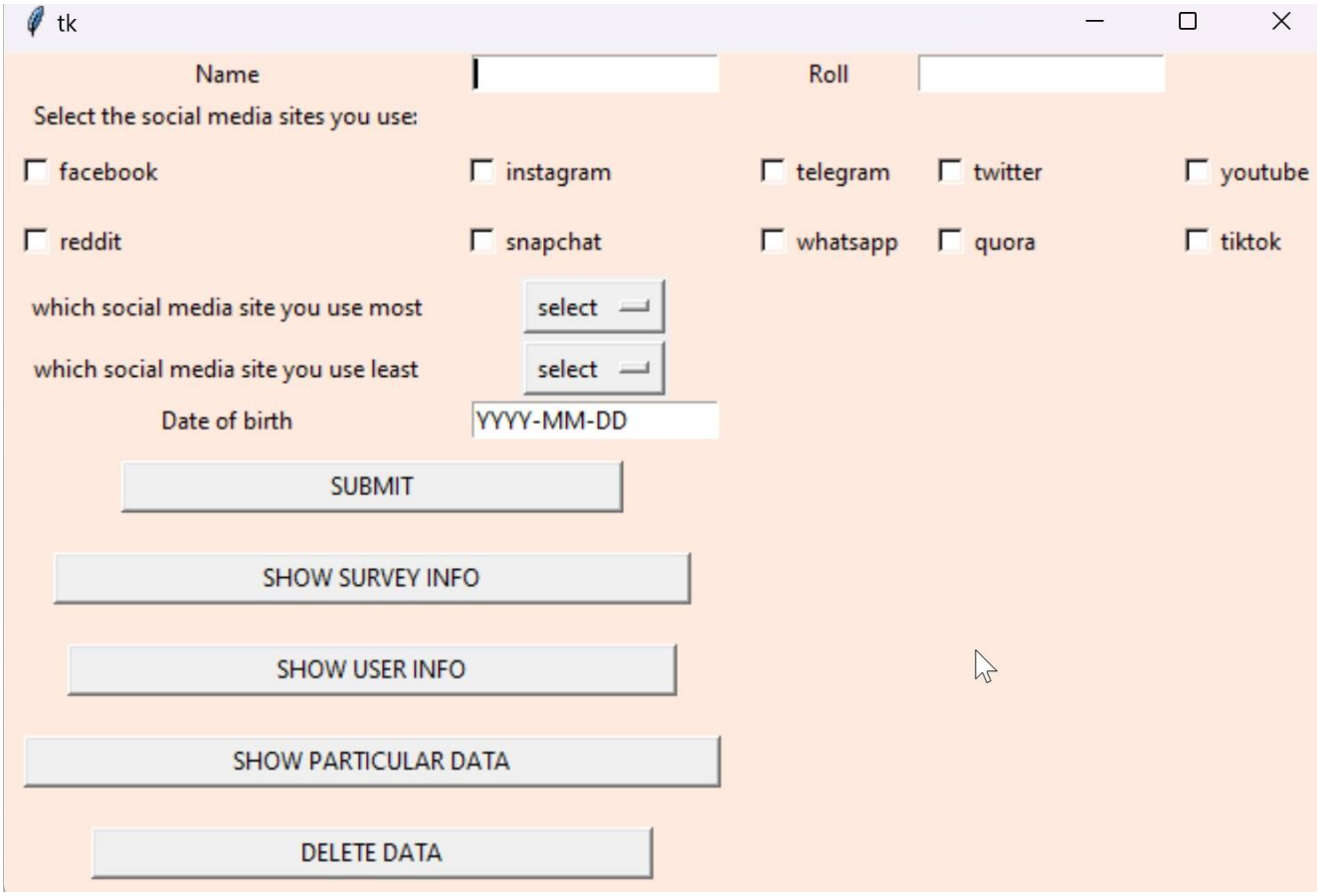

Button(root, text='SHOW USER INFO', command=show_user_info).grid(row=17, column=0, columnspan=2, pady=10, padx=10, ipadx=100)

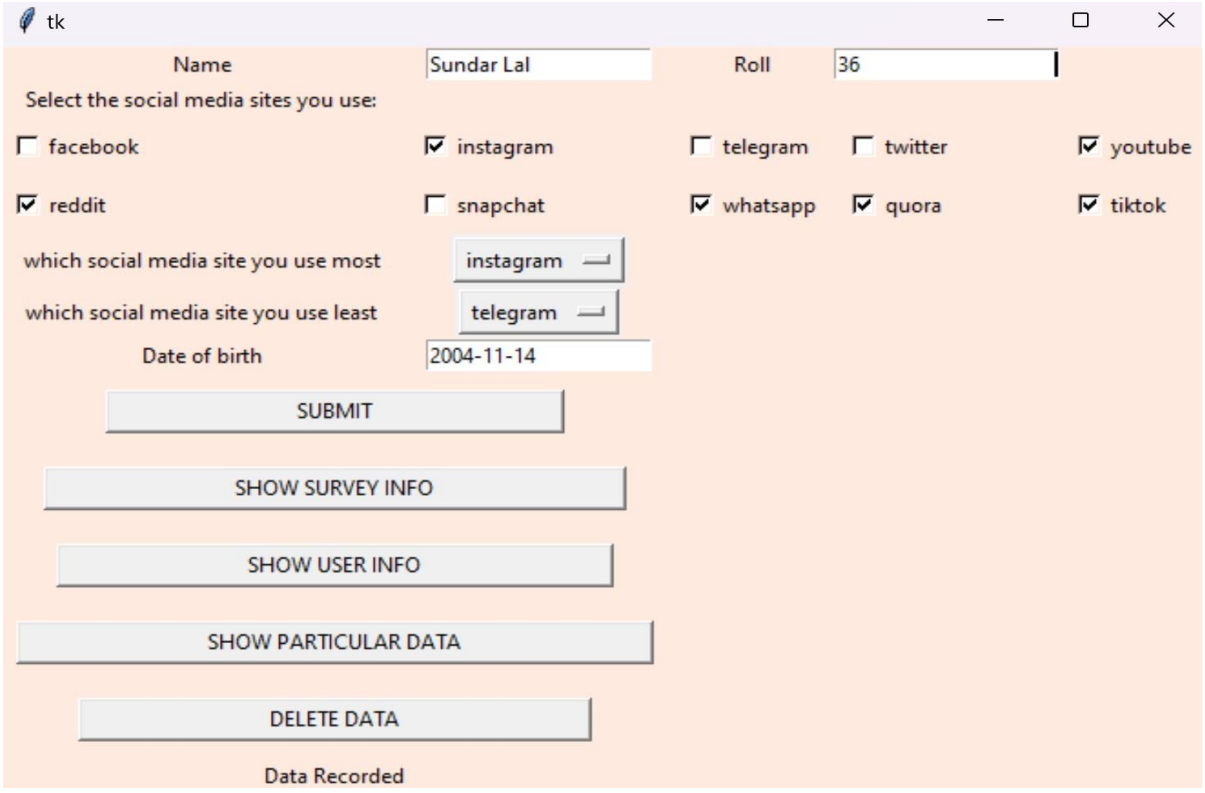

Button(root, text='SHOW PARTICULAR DATA', command=show_particular_data_window).grid(row=18, column=0, columnspan=2, pady=10, padx=10, ipadx=100)

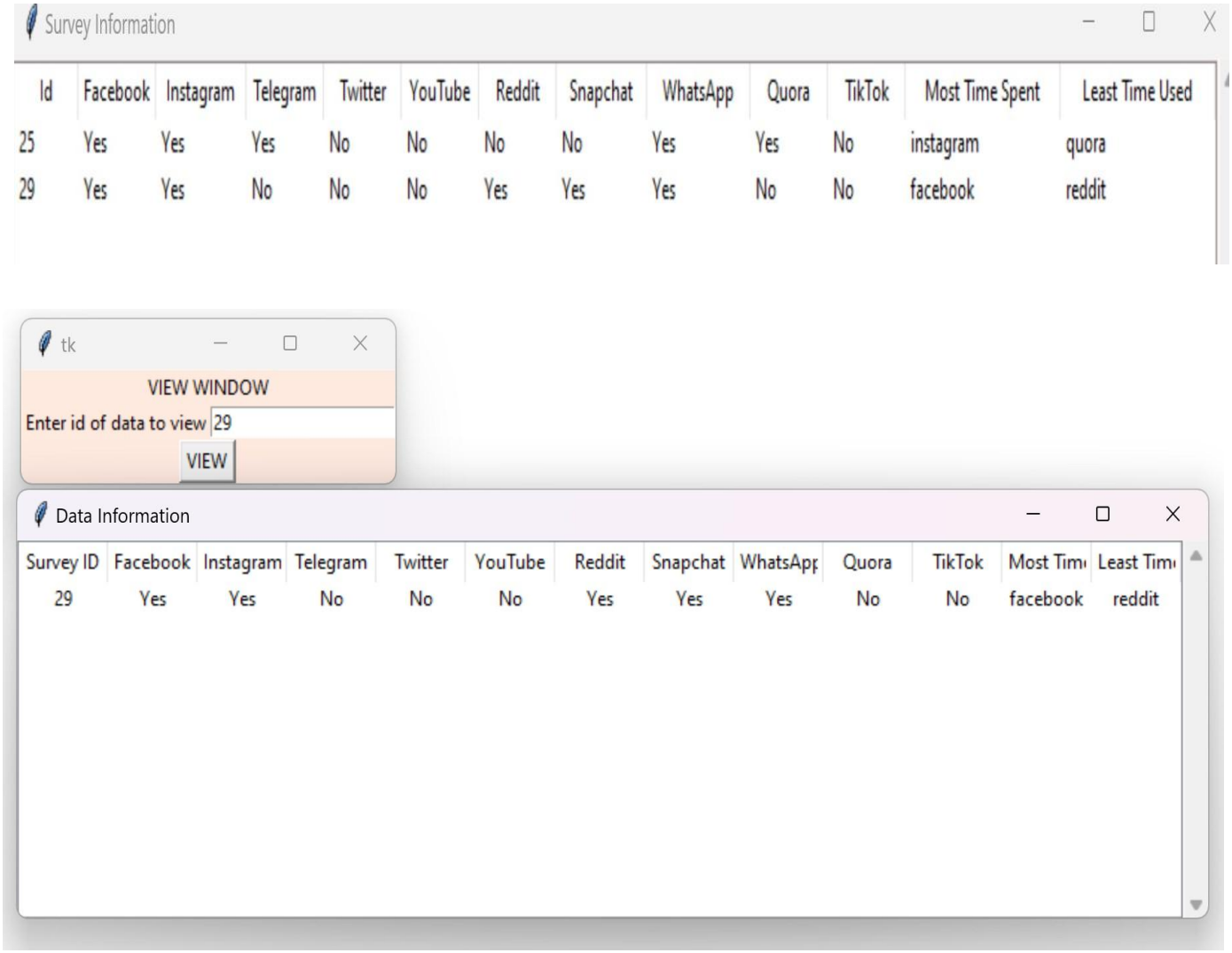

Button(root, text='DELETE DATA', command=delete_data_window).grid(row=19, column=0, columnspan=2, pady=10, padx=10, ipadx=100)

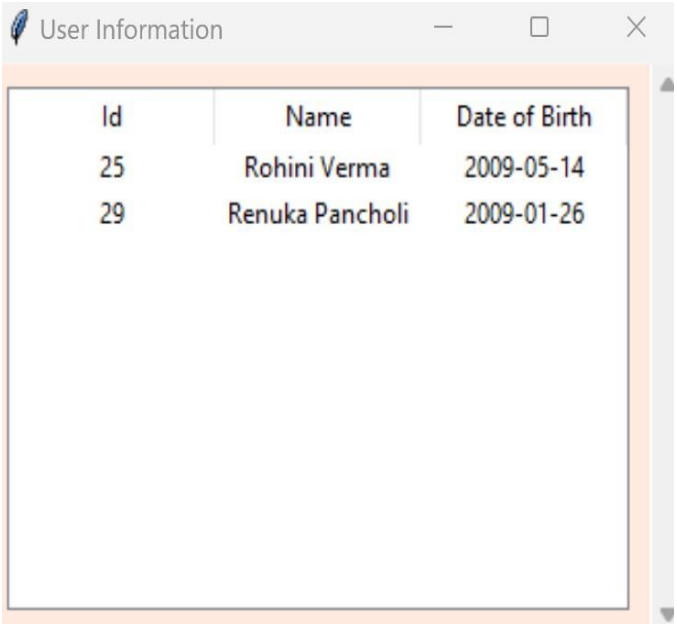

root.mainloop()
```

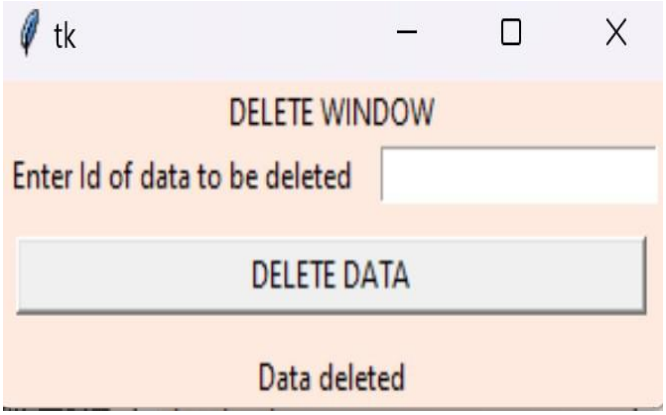
Output


tk
Name
Roll
Select the social media sites you use:
facebook
instagram
telegram
twitter
youtube
reddit
snapchat
whatsapp
quora
tiktok
which social media site you use most
select
which social media site you use least
select
Date of birth
YYYY-MM-DD
SUBMIT
SHOW SURVEY INFO
SHOW USER INFO
SHOW PARTICULAR DATA
DELETE DATA


tk
Name
Sundar Lal
Roll
36
Select the social media sites you use:
facebook
instagram
telegram
twitter
youtube
reddit
snapchat
whatsapp
quora
tiktok
which social media site you use most
instagram
which social media site you use least
telegram
Date of birth
2004-11-14
SUBMIT
SHOW SURVEY INFO
SHOW USER INFO
SHOW PARTICULAR DATA
DELETE DATA
Data Recorded


Survey Information
Id Facebook Instagram Telegram Twitter YouTube Reddit Snapchat WhatsApp Quora TikTok Most Time Spent Least Time Used
25 Yes Yes Yes No No No No Yes Yes No instagram quora
29 Yes Yes No No No Yes Yes Yes No No facebook reddit
tk
VIEW WINDOW
Enter id of data to view 29
VIEW
Data Information
Survey ID Facebook Instagram Telegram Twitter YouTube Reddit Snapchat WhatsApp Quora TikTok Most Tim Least Tim
29 Yes Yes No No No Yes Yes Yes No No facebook reddit


User Information
Id Name Date of Birth
25 Rohini Verma 2009-05-14
29 Renuka Pancholi 2009-01-26


tk
DELETE WINDOW
Enter Id of data to be deleted
DELETE DATA
Data deleted

Technological requirements: Hardware & Software

Software

1. Python
2. VS Code
3. MySQL

Hardware

1. Laptop/PC

References

1. https://www.python.org/
2. https://realpython.com/python-gui-tkinter/
3. https://www.tutorialspoint.com/python/python_gui_programming.htm
4. https://codewithap.com/mysql-crud-operations-python/